

HALIFAX DOCKYARD, NS, Canada

WMO: 713280

Lat: 44.660N

Lon: 63.580W

Elev: 12

StdP: 14.69

Time Zone: -4.00 (NAA)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.5	11.1	-2.0	4.9	10.4	1.0	5.8	14.5	34.8	32.9	27.8	30.1	9.5	270	0.569

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.9	81.0	66.5	78.0	65.3	75.2	63.7	69.9	76.9	68.3	74.1	66.8	71.9	7.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.8	102.2	72.4	66.3	96.8	70.8	64.7	91.4	69.5	33.9	76.7	32.5	73.9	31.4	72.1	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.9	21.5	18.7	DB	2.0	87.6	3.1	3.4	-0.3	90.0	-2.1	92.0	-3.9	93.9	-6.1	96.3	(4)
(5)			WB	1.1	72.9	2.9	1.4	-0.9	73.9	-2.6	74.7	-4.2	75.5	-6.3	76.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	47.1	28.0	27.8	32.8	41.7	49.8	58.3	66.8	67.5	61.9	52.8	43.0	33.5	(6)
(7)		DBStd	15.51	9.30	8.19	7.36	5.70	5.51	5.57	4.62	3.89	5.15	6.45	7.22	8.68	(7)
(8)		HDD50	2952	682	620	536	257	70	3	0	0	0	44	229	511	(8)
(9)		HDD65	6756	1146	1040	999	700	472	212	33	20	119	380	659	976	(9)
(10)		CDD50	1894	0	0	2	6	64	251	520	542	357	131	20	1	(10)
(11)		CDD65	223	0	0	0	0	1	10	88	97	25	2	0	0	(11)
(12)		CDH74	869	0	0	4	1	21	77	399	306	56	5	0	0	(12)
(13)		CDH80	141	0	0	0	0	4	16	81	37	3	0	0	0	(13)
(14)	Wind	WSAvg	8.5	8.8	9.5	9.6	9.5	8.7	8.0	6.5	6.7	7.5	8.6	9.2	9.1	(14)
(15)	Precipitation	PrecAvg	55.7	5.2	4.5	4.8	4.3	4.4	4.1	4.0	3.8	3.8	5.0	5.5	5.9	(15)
(16)		PrecMax	70.7	10.7	9.0	9.6	8.2	8.4	13.7	9.0	12.6	8.9	10.1	10.2	10.0	(16)
(17)		PrecMin	38.7	1.7	1.3	1.8	0.7	1.0	1.3	0.5	1.0	1.1	1.3	1.5	2.6	(17)
(18)		PrecStd	7.8	1.9	1.9	1.8	1.7	1.8	2.2	1.9	2.4	1.8	2.0	2.2	2.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.3	50.9	58.2	63.8	76.7	81.6	87.2	84.3	79.5	72.9	62.8	56.9	(19)
(20)			MCWB	51.8	48.6	48.7	50.9	59.2	63.4	69.6	69.3	66.8	61.3	59.6	54.7	(20)
(21)		2%	DB	48.5	45.8	49.5	57.2	67.6	75.3	81.7	80.1	75.1	68.1	59.8	52.8	(21)
(22)			MCWB	46.4	43.3	43.0	46.6	54.6	61.2	67.2	66.7	64.9	61.2	56.7	51.0	(22)
(23)		5%	DB	44.8	42.2	45.6	53.5	62.4	71.4	78.5	77.4	72.1	64.6	56.9	49.2	(23)
(24)			MCWB	42.6	39.8	40.9	46.4	52.0	59.1	65.7	65.0	63.6	59.9	54.0	47.4	(24)
(25)		10%	DB	41.1	39.0	42.4	50.1	58.5	67.6	75.1	74.7	69.5	62.1	54.2	46.0	(25)
(26)			MCWB	38.8	36.6	38.1	44.4	50.1	58.1	64.1	64.0	63.2	57.9	51.3	44.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.9	49.4	51.1	55.1	62.8	67.4	72.3	71.9	70.5	66.9	60.6	55.4	(27)
(28)			MCD B	53.3	50.4	56.3	59.6	72.6	76.2	80.0	79.1	75.4	69.9	62.3	56.7	(28)
(29)		2%	WB	47.0	44.0	45.0	51.1	58.1	64.2	70.1	69.8	68.1	63.3	57.5	51.0	(29)
(30)			MCD B	48.3	45.2	47.9	53.8	63.6	71.9	77.9	76.0	72.1	66.0	59.3	52.3	(30)
(31)		5%	WB	43.0	40.2	41.7	47.9	54.6	61.8	68.3	68.2	66.2	60.8	54.6	47.4	(31)
(32)			MCD B	44.6	41.9	44.7	51.3	59.8	67.5	75.3	73.5	70.0	63.5	56.4	48.9	(32)
(33)		10%	WB	39.0	37.0	38.7	45.3	51.6	59.7	66.6	66.7	64.3	58.1	51.7	43.8	(33)
(34)			MCD B	40.8	38.7	41.8	48.7	56.5	65.0	72.3	71.4	67.7	61.7	53.8	45.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	12.6	11.9	11.9	12.1	13.1	13.0	13.6	12.9	12.3	10.9	10.7	11.4	(35)
(36)			MCDBR	16.7	15.4	16.5	18.2	21.2	21.4	19.3	17.1	15.4	13.7	13.8	15.4	(36)
(37)			MCWBR	16.9	14.9	12.9	11.7	11.5	9.9	8.2	7.3	8.5	9.5	12.4	15.8	(37)
(38)		5% WB	MCD BR	16.7	15.6	15.7	15.3	18.5	17.4	16.2	14.1	12.8	12.4	13.3	14.8	(38)
(39)			MCWBR	17.4	15.5	14.4	12.6	12.5	10.0	9.1	7.8	9.3	10.9	13.6	16.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.304	0.297	0.325	0.350	0.371	0.401	0.417	0.398	0.355	0.333	0.317	0.297	(40)	
(41)		taud	2.464	2.517	2.470	2.443	2.430	2.373	2.350	2.400	2.492	2.551	2.549	2.533	(41)	
(42)		Ebn at Noon	259	286	290	290	285	276	270	271	276	268	254	252	(42)	
(43)		Edh at Noon	23	25	30	34	36	38	38	35	30	24	21	19	(43)	
(44)	All-Sky Solar Radiation	RadAvg	455	738	1116	1365	1569	1707	1699	1584	1288	835	479	358	(44)	
(45)		RadStd	35	55	76	113	200	159	124	110	86	87	51	27	(45)	

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days				
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (33 neighbors)	N/A	N/A	N/A	N/A	+0.65	+0.90	N/A	N/A	+90	+25	(47)

Nomenclature: See separate page